

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L5
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Flow

- Announcement
- Recap
- Module-1 continues

Announcement

- How was the first tutorial?

- Quality of questions?

- Too easy → 7
 - Too difficult → 0
 - Balanced → 30

- ~~Check Moodle regularly~~
(~~courses.iitm.ac.in~~)

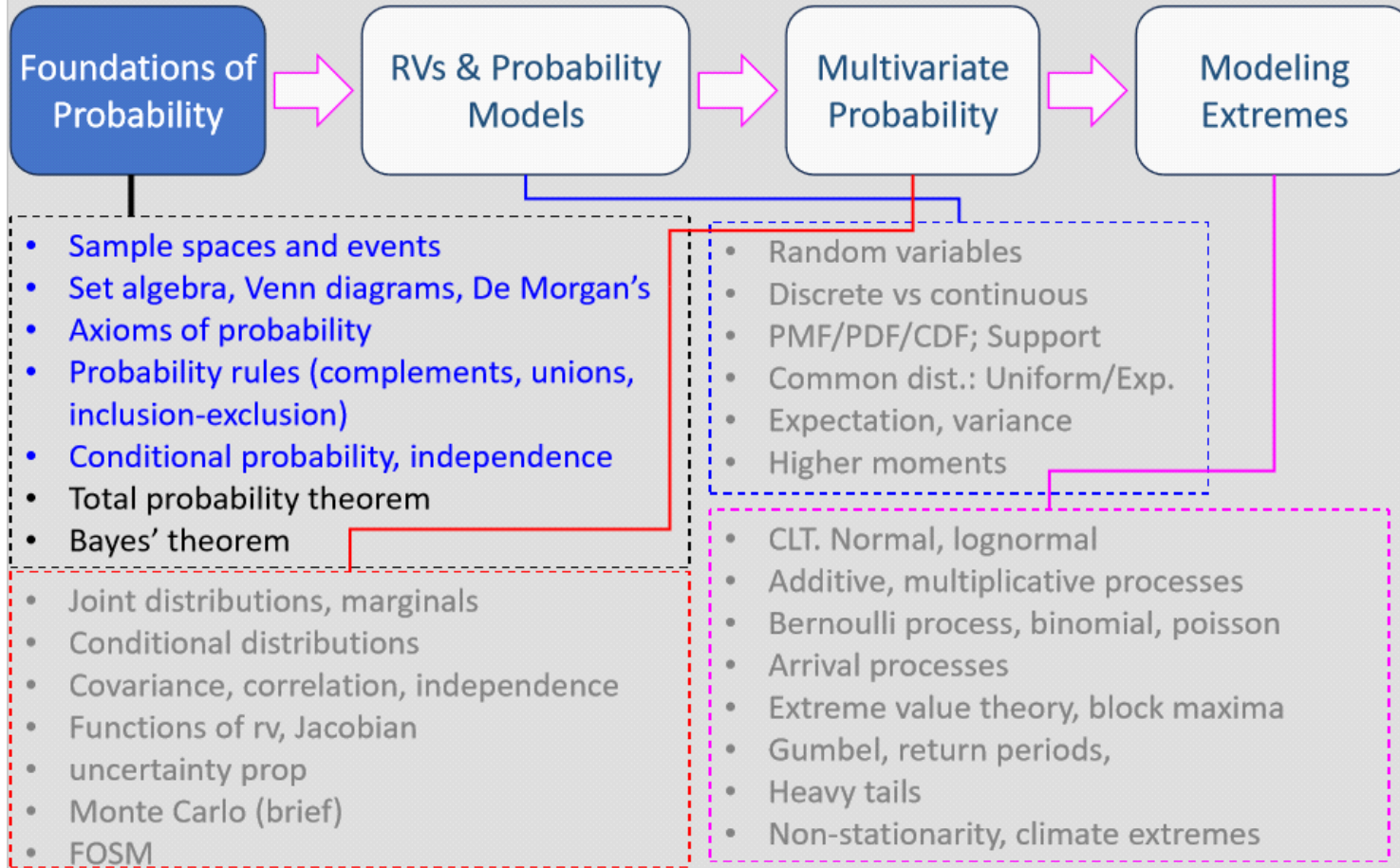
- Course page moved to
GitHub

[https://github.com/prakashvs613/
2026-CV2090-IITM/](https://github.com/prakashvs613/2026-CV2090-IITM/)



Module I: Chapter 1 (M1C1)

Module 1: **Probability** – Understanding Uncertainty



Messages from the Last Class

- **Message 1:** Conditional probability

→ $\Pr(A|B) = \Pr(A \cap B) / \Pr(B)$

→ Narrowing down of Sample space from S to B .



- **Message 2:** Statistical independence

→ IBB $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$

→ Other independence

→ Causal

→ Physical

→ spurious correlation

→ Correlation does NOT imply Causation.

- **Message 3:** Multiplication rule

$$\Pr(E_1 E_2 \cdots E_n) = \Pr(E_1 | E_2 E_3 \cdots E_n) \cdot \Pr(E_2 | E_3 E_4 \cdots E_n) \cdots \Pr(E_{n-1} | E_n) \cdot \Pr(E_n)$$

conditionals

marginale

→ Conditionals, Marginal

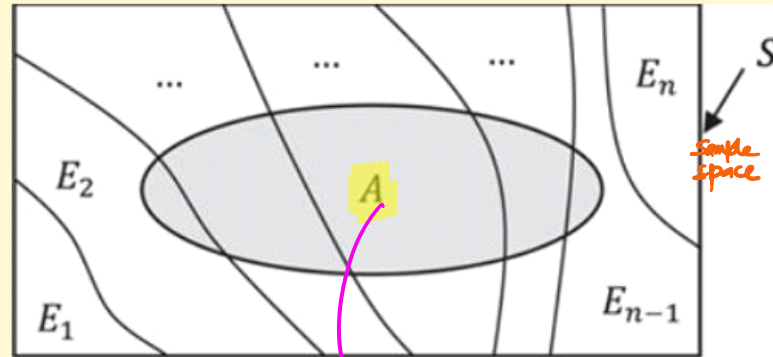
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Probability rules: total probability rule

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- If $E_1, E_2, \dots, E_n$ are n **MECE** events,
partitioning a complex system into $E_1, E_2, \dots, E_n$



$$Pr(A) = Pr(AE_1) + Pr(AE_2) + \dots + Pr(AE_n)$$

$$= \sum_{i=1}^n Pr(AE_i)$$

$$= \sum_{i=1}^n Pr(A|E_i) \cdot Pr(E_i)$$

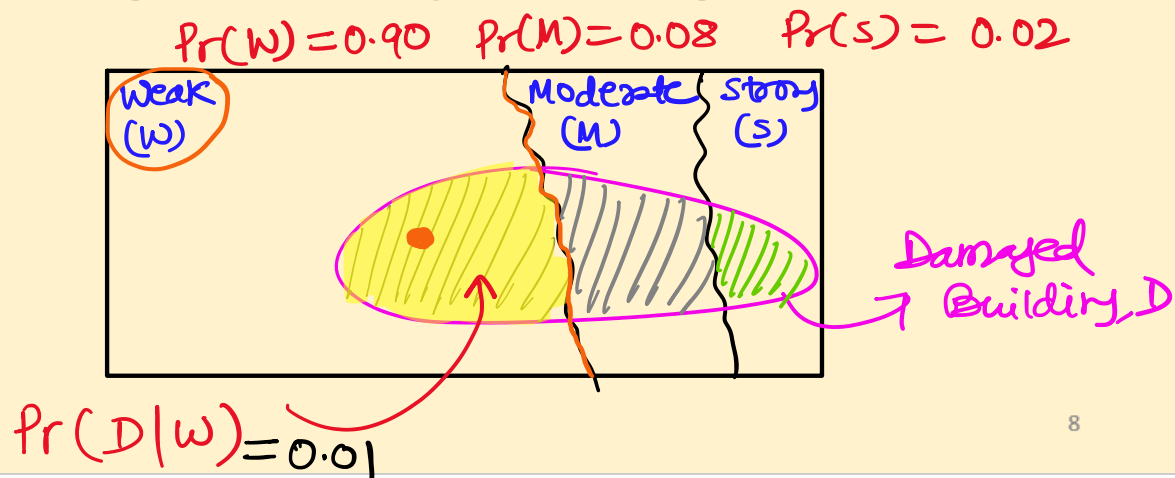
→ Break down $Pr(A)$ into conditional probabilities
 → clever selection of E_i 's is crucial for calculating $Pr(A)$.

Probability rules: total probability rule

The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively.

Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

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Probability rules: total probability rule

$$Pr(D) = \left\{ \frac{Pr(D|W)}{Pr(W)} \right\} + \left\{ \frac{Pr(D|M)}{Pr(M)} \right\} + \left\{ \frac{Pr(D|S)}{Pr(S)} \right\}$$

$$= Pr(DW) + Pr(DM) + Pr(DS)$$

$$= \frac{0.01}{\times 0.90} + \frac{0.10}{\times 0.08} + \frac{0.60}{\times 0.02}$$

$$= 0.9\% + 0.8\% + 1.2\%$$

$$= 2.9\%$$

Probability of
damage

$Pr(\text{Shaking})$

→ Seismologist

$Pr(\text{Damage} | \text{Shaking})$

→ Structural Engineer

Total Probability Rule: Behind the Scenes!

→ Divide and Conquer!

Partitioning Complex ~~Structural~~ Systems

probability that Chennai

Airport is closed tomorrow?

Weather Prediction (Rains)

- + No Rain
- + Minor Rain
- + Major Rain
- + Cyclone

Probability rules: Bayes' rule (motivation)

- Covid tests: Why even a 99% accurate test result often means only a 10% actual chance of having the disease?
- Suppose
 - Prevalence = 1%
 - Sensitivity = 99% (true positive)

Probability rules: Bayes' rule (motivation)

Surfside condominium collapse (June 24, 2021)

12-storied, 98 casualties

Tentative settlement, valued at \$997 million, reached with families of victims of Surfside condominium collapse



Probability rules: Bayes' rule (motivation)

Surfside condominium collapse

23 languages

Article Talk

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From Wikipedia, the free encyclopedia

Coordinates: 25°52′23″N 80°07′15″W

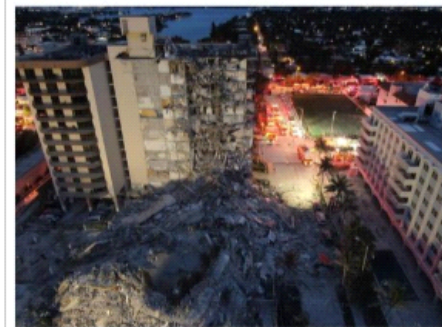
"Miami building collapse" redirects here. For other incidents of building collapses in or near Miami, see [List of building and structure collapses](#).

On June 24, 2021, at approximately 1:22 a.m. EDT, Champlain Towers South, a 12-story beachfront condominium in the Miami suburb of Surfside, Florida, United States, partially collapsed, causing the deaths of 98 people.^[4] Four people were rescued from the rubble, but one of them died of injuries shortly after arriving at the hospital.^[5] Eleven others were injured.^[6] Approximately 35 were rescued the same day from the intact portion of the building,^[7] which was demolished ten days later.

A contributing factor under investigation was long-term degradation of reinforced concrete structural support in the basement-level parking garage under the pool deck, due to water penetration and corrosion of the reinforcing steel. The problems had been reported in 2018 and noted as "much worse" in April 2021. A \$15 million program of remedial works had been approved before the collapse, but the main structural work had not started. Other possible factors considered included land subsidence, insufficient reinforcing steel, and corruption during construction.^{[8][9][10]}

In June 2026, the National Institute of Standards and Technology (NIST) released their technical

Surfside condominium collapse



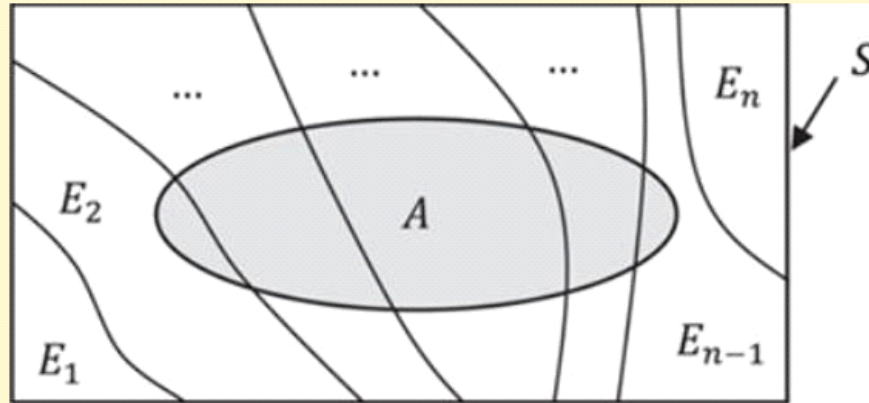
View of Champlain Towers South site, the morning of the collapse

→ Misinterpretation of structural damage due to a lack of "update model of structural Deterioration"

Probability rules: Bayes' rule

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- If $E_1, E_2, \dots, E_n$ are n MECE events,



Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

Next Class

- Random variables
 - Discrete, continuous
 - PMF/PDF/CDF

Thank you!